Web-Based Subscription Delivery System with Route Optimization

Project Members

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Problem Statement

To design a web-based subscription delivery system that optimally assigns and routes delivery partners from a single depot to multiple customers within fixed time slots using clustering and routing algorithms.

Project Description

The system is a web-based subscription delivery platform for fruit bowls where customers can register, choose subscription plans, customize orders, and select fixed morning or evening delivery slots. All deliveries start from a single pickup location. Customers can pause, resume, or skip deliveries before a defined cut-off time, after which orders are locked. The system automatically assigns orders to delivery partners and generates optimized delivery routes using clustering and routing algorithms. Each partner collects all items from the depot, delivers to multiple customers within the time slot, and returns to the depot. The platform provides separate dashboards for admins, delivery partners, and customers, and is designed to be secure, scalable, and reliable.

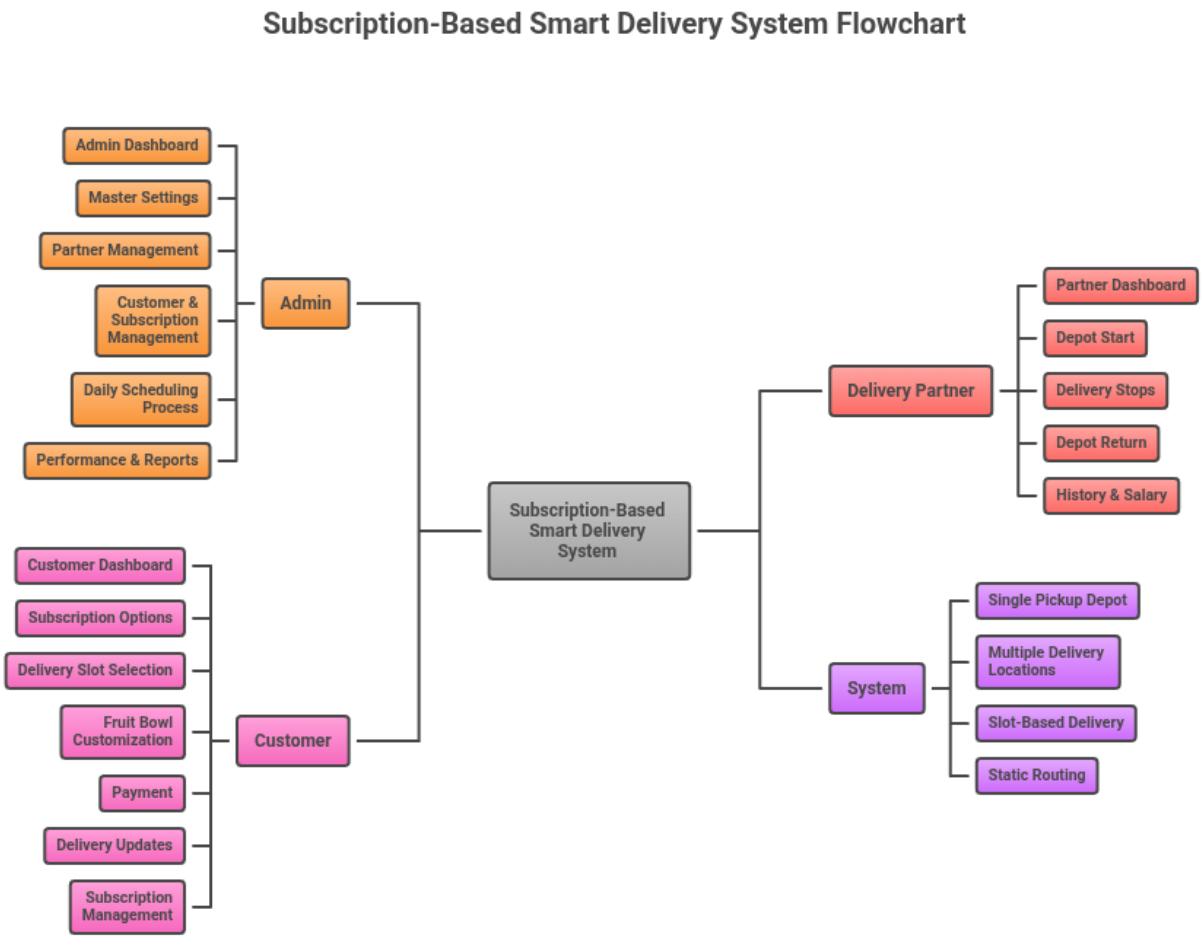
Actors

- Admin
- Customer
- Delivery Partner

System Architecture

- **Web-based application**
- **Centralized backend**
- **Role-based access control**
- **Database-driven scheduling**
- **Static optimization algorithms**

Overall Workflow



Authentication & Role Management

Pre-condition :-- User not logged in

Process :-- Login → Role identification → Session/JWT generation → Dashboard redirection

Output Sec :-- ure login and role-based dashboard access

Customer Registration & Subscription Management

Pre-condition :-- New customer or new subscription

Process :-- Enter personal and delivery details → Choose subscription plan and delivery slot → Customize fruit bowl → Complete payment

Output :-- Customer account created, subscription activated, orders scheduled.

Order Generation and Cutoff Managment

Pre-condition :-- Active subscriptions exist and cut-off time not reached

Process :-- Daily order generation → Apply pause/skip rules → Group orders by delivery slot → Lock orders after cut-off

Output :-- Finalized orders ready for scheduling with no further modifications

Delivery Partner Management

Pre-condition :-- Admin logged in

Process :-- Add or remove delivery partners → Set partner availability per slot → Activate or deactivate accounts

Output :-- Updated partner availability stored

Delivery Scheduling & Customer Clustering

Pre-condition :-- Orders locked and delivery partners available

Process :-- Fetch customer locations → Apply K-Means clustering → Assign each cluster to a delivery partner

Output :-- Balanced customer clusters created per partner

Route Optimization

Pre-condition :-- Customer clusters available

Process :-- Start route from depot → Apply Nearest Neighbor TSP → Optimize delivery sequence → Calculate route time

Output :-- Optimized delivery routes generated

Schedule Validation & Admin Approval

Pre-condition :-- Routes generated

Process :-- Validate route duration against slot time → Flag overloaded routes → Admin approval or manual override

Output :-- Final delivery schedules locked for execution

Delivery Execution

Pre-condition :-- Schedule approved and delivery slot started

Process :-- Delivery partner views assigned route → Delivers orders in sequence → Marks deliveries as completed

Output :-- Orders marked as delivered and logs recorded

Salary & Earnings Calculation

Pre-condition :-- Delivery completion data available

Process :-- Count completed deliveries → Calculate monthly salary → Apply bonuses or penalties

Output :-- Earnings updated and salary reports generated

Admin Monitoring & Reporting

Pre-condition :-- Admin logged in

Process :-- Aggregate delivery, revenue, and partner performance data → Generate reports

Output :-- Dashboards and downloadable reports available

Assumptions

- **Single depot location**
- **Static travel conditions**
- **Fixed delivery slots**
- **Fixed delivery time per stop**